

1. In class we have seen how to determine the superficial degree of divergence of general field theories with fermions, scalars and spin-1 particles in four dimensional space-time. Repeat the analysis and find an expression for the superficial degree of divergence for these theories in d dimensional space-times. What are the conditions for renormalizability? Apply your analysis to catalogue the type of diagrams which are superficially divergent in a theory with an interaction lagrangian of the type $g\phi^3$ in 6 dimensional space-time.
2. Consider the renormalization of a scalar field theory defined by the lagrangian in 6 dimensional space-times:

$$L = \frac{1}{2}Z_2\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}Z_2Z_m m^2\phi^2 - Z_1\frac{g}{3!}\phi^3,$$

where, the Z_i are the renormalization constants. By splitting the lagrangian into $L_{int} + L_{c.t.}$ as discussed in class, determine the renormalization constants to one loop order. Use dimensional regularization.